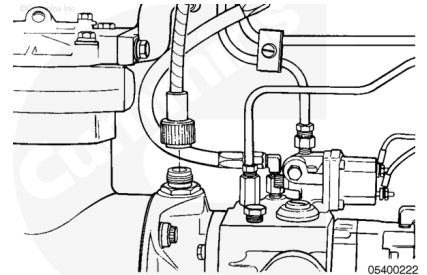


Trainings ▾

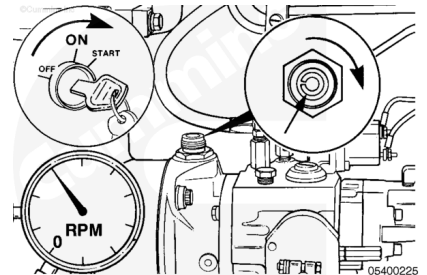
Rotation Check

Remove the tachometer drive cable from the fuel pump.



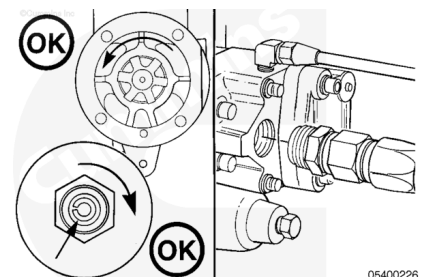
Crank the engine and make sure the tachometer drive shaft is turning.

If the tachometer drive shaft is **not** turning, remove the fuel pump and check the air compressor and accessory drive operation.



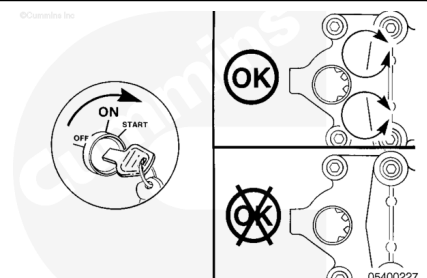
If the tachometer drive shaft rotates, the gear pump rotation **must** be checked.

Remove the fuel supply hose at the gear pump.



Look into the gear pump and crank the engine. The gear pump gears **must** turn.

If the gear pump gears do **not** turn, replace the fuel pump.



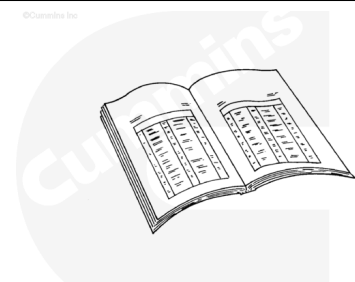
Preparatory Steps

All Applications Except Rail

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

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ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) cable last.

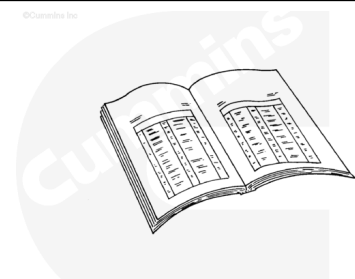
- Clean the fuel pump and surrounding area.
- Disconnect the battery cables.

Rail Applications

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

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⚠ WARNING ⚠

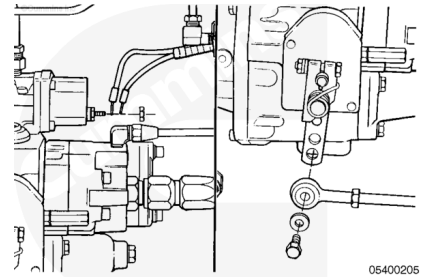
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) cable last.

- Clean the fuel pump and surrounding area.
- Disconnect the battery cables.
- Remove the lubricating oil scavenge pump.

Remove

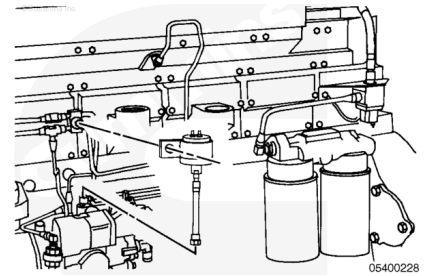
Remove the wire to the fuel shutoff valve.

Remove the linkage from the throttle lever.



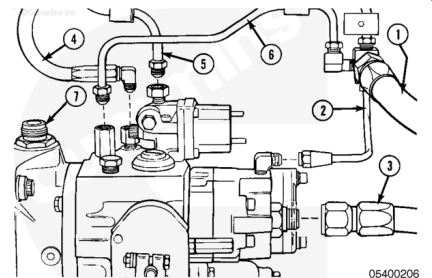
Note : This step applies only to engines equipped with STC.

Remove the STC fuel rail switch.



Remove the below listed items:

- Fuel drain (1)
- Gear pump cooling drain (2)
- Gear pump suction line (3)
- AFC supply hose (4)
- Fuel supply to injectors (5)
- AFC drain hose (6)
- Tachometer cable (7).

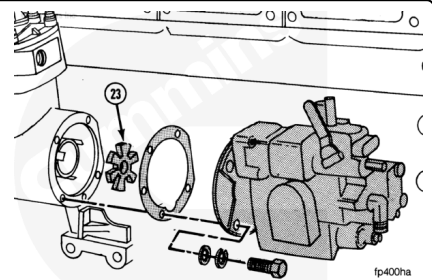


Remove the four fuel pump mounting capscrews.

Remove the fuel pump.

Remove the drive coupling (23).

Remove and discard the gasket.

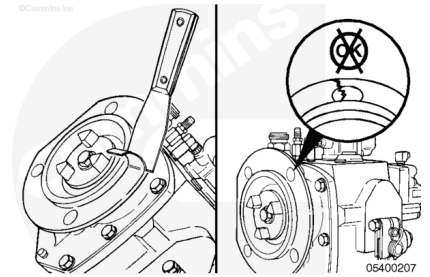


Clean and Inspect for Reuse

Clean the fuel pump and the air compressor or accessory drive mounting surfaces.

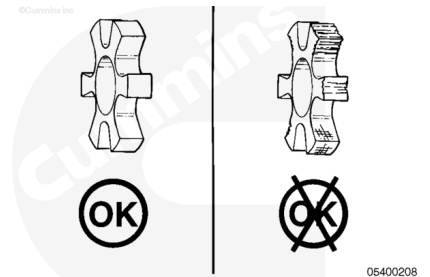
Inspect the mounting surfaces for damage.

If the mounting surface is damaged, the fuel pump **must** be repaired or replaced.



Inspect the spider coupling for damage.

If the spider coupling is damaged, it **must** be replaced.



Install

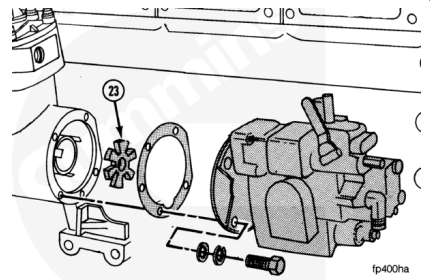
All K19 engines use a white nylon or light green fuel pump drive coupling.

Install the drive coupling (23).

Install the gasket, fuel pump and four capscrews.

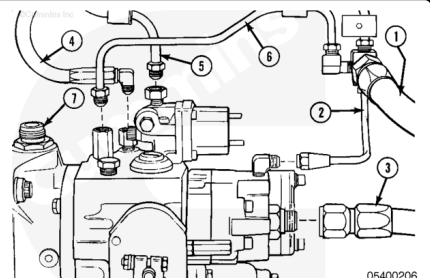
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the following items:

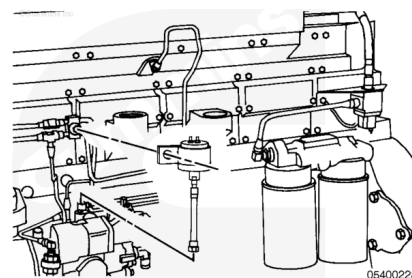
- Fuel drain (1)
- Gear pump cooling drain (2)
- Gear pump suction line (3)
- AFC fuel drain (4)
- Fuel supply to the injectors (5)
- AFC air supply hose (6)
- Tachometer cable (7).



Note : This step applies **only** to engines equipped with **STC**.

Install the STC pressure switch.

Connect the hose to the switch and to the fuel pressure tube fitting on the fuel pump.

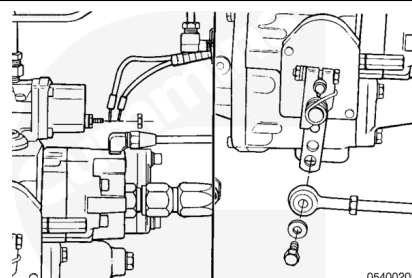


Make sure the wire connection nut and ground post are clean.

Install the electric wire to the fuel shutoff valve.

Tighten the nut.

Install the linkage to the throttle lever.



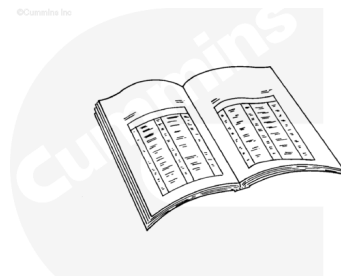
Finishing Steps

All Applications Except Rail

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- Connect the battery cables.

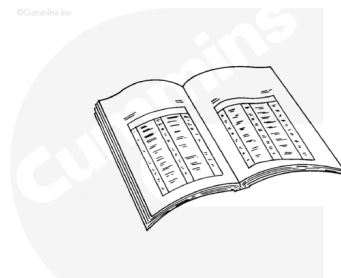


Rail Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) cable last.

- Connect the battery cables.
- Install the lubricating oil scavenge pump.



Last Modified: 03-May-2019
